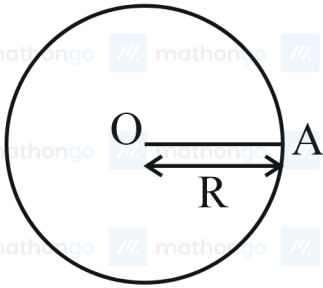
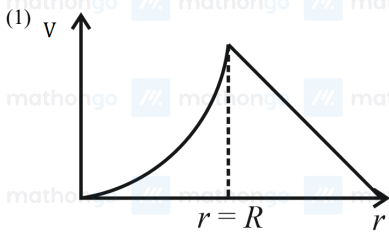


1. For a uniformly charged thin spherical shell, the electric potential (V) radially away from the centre (O) of shell can be graphically represented as



[2023 (06 Apr Shift 1)]



2. A dipole comprises of two charged particles of identical magnitude q and opposite in nature. The mass m of the positive charged particle is half of the mass of the negative charged particle. The two charges are separated by a distance l . If the dipole is placed in a uniform electric field $\vec{E}$; in such a way that dipole axis makes a very small angle with the electric field, $\vec{E}$. The angular frequency of the oscillations of the dipole when released is given by:

[2023 (06 Apr Shift 2)]

- (1) $\sqrt{\frac{3qE}{2ml}}$
 (2) $\sqrt{\frac{8qE}{ml}}$
 (3) $\sqrt{\frac{4qE}{ml}}$
 (4) $\sqrt{\frac{8qE}{3ml}}$

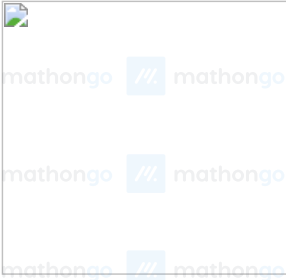
3. Experimentally it is found that 12.8 eV energy is required to separate a hydrogen atom into a proton and an electron. So the orbital radius of the electron in a hydrogen atom is $\frac{9}{x} \times 10^{-10}$ m.

The value of the x is: _____.

(1 eV = 1.6×10^{-19} J, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$ and electronic charge = 1.6×10^{-19} C)

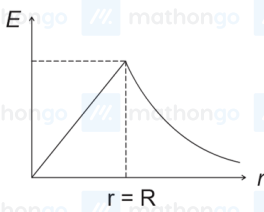
[2023 (06 Apr Shift 2)]

4. Graphical variation of electric field due to a uniformly charged insulating solid sphere of radius R , with distance r from the centre O is represented by:

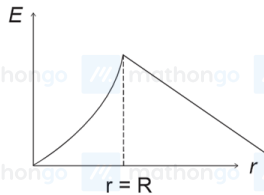


[2023 (08 Apr Shift 1)]

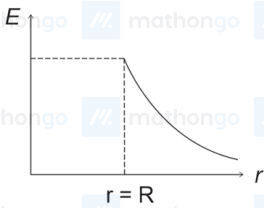
(1) Dig



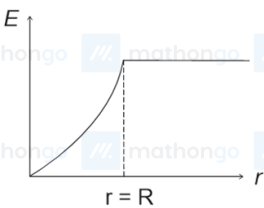
(2)



(3)



(4)



5. An electric dipole of dipole moment is $6.0 \times 10^{-6} \text{ C m}$ placed in a uniform electric field of $1.5 \times 10^3 \text{ N C}^{-1}$ in such a way that dipole moment is along electric field. The work done in rotating dipole by 180° in this field will be _____ mJ.

[2023 (08 Apr Shift 1)]

6. Electric potential at a point P due to a point charge of $5 \times 10^{-9} \text{ C}$ is 50 V. The distance of P from the point charge is:

(Assume, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$)

[2023 (08 Apr Shift 2)]

(1) 9 cm

(2) 3 cm

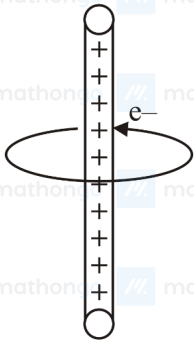
(3) 0.9 cm

(4) 90 cm

7. Three concentric spherical metallic shells X , Y and Z of radius a , b and c respectively [$a < b < c$] have surface charge densities σ , $-\sigma$ and σ , respectively. The shells X and Z are at same potential. If the radii of X & Y are 2 cm and 3 cm, respectively. The radius of shell Z is _____ cm.

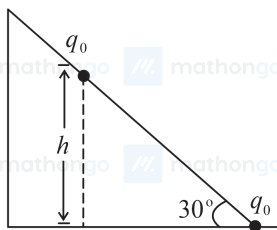
[2023 (10 Apr Shift 1)]

8. An electron revolves around an infinite cylindrical wire having uniform linear charge density $2 \times 10^{-8} \text{ C m}^{-1}$ in circular path under the influence of attractive electrostatic field as shown in the figure. The velocity of electron with which it is revolving is _____ $\times 10^6 \text{ m s}^{-1}$. Given mass of electron $= 9 \times 10^{-31} \text{ kg}$



[2023 (10 Apr Shift 2)]

9. As shown in the figure, a configuration of two equal point charges ($q_0 = +2\mu\text{C}$) is placed on an inclined plane. Mass of each point charge is 20 g. Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height $h = x \times 10^{-3} \text{ m}$. The value of x is (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$, $g = 10 \text{ m s}^{-2}$)



[2023 (11 Apr Shift 1)]

10. If V is the gravitational potential due to sphere of uniform density on its surface, then its value at the centre of sphere will be:

[2023 (11 Apr Shift 2)]

- (1) $\frac{4}{3} V$
- (2) $\frac{V}{2}$
- (3) V
- (4) $\frac{3V}{2}$

11. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : If an electric dipole of dipole moment $30 \times 10^{-5} \text{ C m}$ is enclosed by a closed surface, the net flux coming out of the surface will be zero.

Reason R : Electric dipole consists of two equal and opposite charges.

In the light of above, statements, choose the correct answer from the options given below.

[2023 (12 Apr Shift 1)]

- (1) Both A and R are true and R is the correct explanation of A
- (2) A is false but R is true
- (3) A is true but R is false
- (4) Both A and R are true and R is NOT the correct explanation of A

12. Two charges of each magnitude 0.01 C and separated by a distance of 0.4 mm constitute an electric dipole. If the dipole is placed in an uniform electric field $\vec{E}$ of $10 \text{ dyne} \cdot \text{C}^{-1}$ making 30° angle with $\vec{E}$, the magnitude of torque acting on dipole is:

[2023 (13 Apr Shift 1)]

- (1) $4.0 \times 10^{-10} \text{ N m}$
- (2) $1.0 \times 10^{-8} \text{ N m}$
- (3) $1.5 \times 10^{-9} \text{ N m}$
- (4) $2.0 \times 10^{-10} \text{ N m}$

13. A thin infinite sheet charge and an infinite line charge of respective charge densities $+\sigma$ and $+\lambda$ are placed parallel at 5 m distance from each other. Points P and Q are at $\frac{3}{\pi} \text{ m}$ and $\frac{4}{\pi} \text{ m}$ perpendicular distances from line charge towards sheet charge, respectively. E_p and E_q are the magnitudes of resultant electric field intensities at point P and Q respectively. If $\frac{E_p}{E_q} = \frac{4}{a}$ for $2|\sigma| = |\lambda|$, then the value of a is _____.

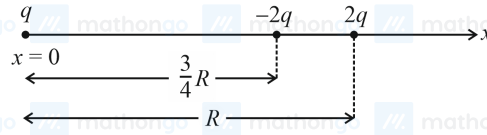
[2023 (13 Apr Shift 1)]

14. A $10\ \mu\text{C}$ charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are: n

[2023 (13 Apr Shift 2)]

- (1) $7\ \mu\text{C}, 3\ \mu\text{C}$
- (2) $8\ \mu\text{C}, 2\ \mu\text{C}$
- (3) $5\ \mu\text{C}, 5\ \mu\text{C}$
- (4) $9\ \mu\text{C}, 1\ \mu\text{C}$

15. Three point charges q , $-2q$ and $2q$ are placed on x axis at a distance $x = 0$, $x = \frac{3}{4}R$ and $x = R$ respectively from origin as shown. If $q = 2 \times 10^{-6}\ \text{C}$ and $R = 2\ \text{cm}$, the magnitude of net force experienced by the charge $-2q$ is ____ N.



[2023 (13 Apr Shift 2)]

16. The electric field due to a short electric dipole at a large distance (r) from center of dipole on the equatorial plane varies with distance as :

[2023 (15 Apr Shift 1)]

- (1) r
- (2) $\frac{1}{r^2}$
- (3) $\frac{1}{r^3}$
- (4) $\frac{1}{r}$

ANSWER KEYS

1. (2) 2. (1) 3. (16) 4. (1) 5. (18) 6. (4) 7. (5) 8. (8)
9. (300) 10. (4) 11. (1) 12. (4) 13. (6) 14. (3) 15. (5440) 16. (3)

1. (2)

The potential is constant inside the sphere as the electric field is zero inside the sphere. So, the value is given by

$$V = \frac{Q}{4\pi\epsilon_0 R} \text{ for } r < R$$

The potential at the surface is

$$V = \frac{Q}{4\pi\epsilon_0 R}$$

The potential outside the sphere starts decreasing and is given by the formula

$$V = \frac{Q}{4\pi\epsilon_0 r} \text{ for } r > R$$

2. (1)

The data given is

l = distance between the charges

q = charge

m = mass of the positive charge

$2m$ = mass of negative charge

The moment of inertia of the system is

$$I = \left(\frac{m \times 2m}{m+2m} \right) (l)^2 = \frac{2ml^2}{3}$$

The angular frequency can be written as

$$\omega = \sqrt{\frac{pE}{I}} \dots (i)$$

Substituting the value of moment of inertia in the equation (i) we get

$$\omega = \sqrt{\frac{pE}{\frac{2ml^2}{3}}} = \sqrt{\frac{3pE}{2ml^2}}$$

The value of the dipole moment is

$$p = ql$$

Hence, the angular frequency becomes

$$\omega = \sqrt{\frac{3qE}{2ml}}$$

This question was given bonus by NTA as none of the options matched in the original paper.

3. (16)

The given data is

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

The formula for the value of energy in a hydrogen atom is given by

$$E = \frac{q^2}{8\pi\epsilon_0 r}$$

$$\Rightarrow E = \frac{q^2}{2(4\pi\epsilon_0)r}$$

$$\Rightarrow r = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{12.8 \times 2}$$

$$\Rightarrow r = \frac{9}{16} \times 10^{-10} \text{ m}$$

It is given that

$$r = \frac{9}{x} \times 10^{-10} \text{ m}$$

$$\Rightarrow \frac{9}{16} \times 10^{-10} = \frac{9}{x} \times 10^{-10}$$

$$\Rightarrow x = 16$$

4. (1)

The electric field inside a solid insulated sphere is given by

$$E_{\text{inside}} = \frac{\rho r}{3\epsilon_0} \text{ for } r < R \text{ where } \rho \text{ is volume charge density.}$$

Hence, the electric field varies linearly with the distance as

$$E_{\text{inside}} \propto r^1$$

Now, the electric field for $r > R$ is given by

$$E_{\text{outside}} = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$E_{\text{outside}} \propto \frac{1}{r^2}$$

Thus, the graph starts decreasing as a power law graph as shown in the figure for the region $r > R$.

Hence, this graph is the right option.

5. (18)

The work done for rotating a dipole is given by the change in the potential energy. The potential energy is given by

$$U = -\vec{p} \cdot \vec{E}$$

$$W = \Delta U = U_f - U_i$$

So, the work done is

$$W = -pE \cos 180^\circ - (-pE \cos 0^\circ) = 2pE$$

$$= 2 \times 6 \times 10^{-6} \times 1.5 \times 10^3 = 18 \text{ mJ}$$

6. (4)

The electric potential for a point charge is given by

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

The data given is

$$V = 50 \text{ V}$$

$$Q = 5 \times 10^{-9} \text{ C}$$

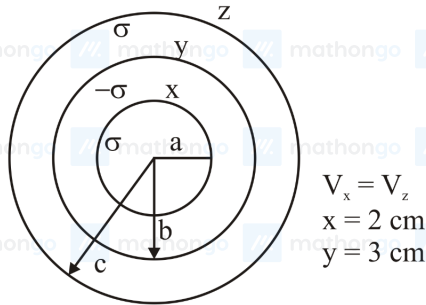
So, the distance of the point is

$$r = \frac{1}{4\pi\epsilon_0} \frac{Q}{V}$$

$$\Rightarrow r = 9 \times 10^9 \times \frac{5 \times 10^{-9}}{50}$$

$$\Rightarrow r = 0.9 \text{ m} = 90 \text{ cm}$$

7. (5)



$$V_x = V_z$$

$$x = 2 \text{ cm}$$

$$y = 3 \text{ cm}$$

The charges on the spheres are

$$q_X = \sigma 4\pi a^2$$

$$q_Y = -\sigma 4\pi b^2$$

$$q_Z = \sigma 4\pi c^2$$

It is given that $V_X = V_Z$

So,

$$\frac{q_X}{4\pi\epsilon_0 a} + \frac{q_Y}{4\pi\epsilon_0 b} + \frac{q_Z}{4\pi\epsilon_0 c} = \frac{q_X}{4\pi\epsilon_0 c} + \frac{q_Y}{4\pi\epsilon_0 c} + \frac{q_Z}{4\pi\epsilon_0 c}$$

$$\Rightarrow \frac{\sigma 4\pi a^2}{a} - \frac{\sigma 4\pi b^2}{b} + \frac{\sigma 4\pi c^2}{c} = \frac{\sigma 4\pi (a^2 - b^2 + c^2)}{c}$$

$$\Rightarrow c(a - b + c) = (a^2 - b^2 + c^2)$$

$$\Rightarrow c = a + b = 2 \text{ cm} + 3 \text{ cm} = 5 \text{ cm}$$

8. (8)

The electric field is given by

$$E = \left(\frac{2k\lambda}{r} \right)$$

The required centripetal force is

$$F = \left(\frac{mv^2}{r} \right)$$

The force due to the field is equal to the required centripetal force,

$$\Rightarrow \left(\frac{2k\lambda}{r} \right) e = \frac{mv^2}{r}$$

$$\Rightarrow v = \sqrt{\frac{2k\lambda e}{m}}$$

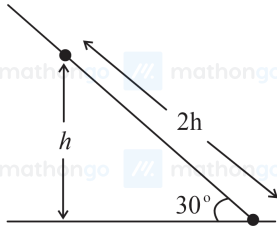
$$= \sqrt{\frac{2 \times 9 \times 10^9 \times 2 \times 10^{-8} \times 1.6 \times 10^{-19}}{9 \times 10^{-31}}}$$

$$= \sqrt{1.6 \times 4 \times 10^{13}}$$

$$= 4 \times 2 \times 10^6 \text{ m s}^{-1}$$

$$= 8 \times 10^6 \text{ m s}^{-1}$$

9. (300)



For the condition of equilibrium, the Coulomb force is equal to $mg \sin \theta$. Hence,

$$\frac{kq_0^2}{4h^2} = (mg \sin \theta)$$

$$\Rightarrow \frac{kq_0^2}{4h^2} = (20 \times 10^{-3}) \times 10 \times \frac{1}{2}$$

$$\Rightarrow \frac{9 \times 10^9 \times 4 \times 10^{-12}}{4h^2} = 20 \times 10^{-3} \times 10 \times \frac{1}{2}$$

$$\Rightarrow h = \left(\sqrt{\frac{9}{100}} \right) \text{ m} = 0.3 \text{ m} = 300 \times 10^{-3} \text{ m}$$

$$h^2 = \left(\frac{9}{100} \right) \Rightarrow h = \left(\frac{3}{10} \right) \text{ m} = 0.3 \text{ m}$$

$$= 300 \times 10^{-3} \text{ m}$$

10. (4)

The magnitude of gravitational potential of a solid sphere at a distance r from the centre is

$$V = \frac{-GM}{2R^3} (3R^2 - r^2)$$

$$\text{At } r = R, V = \frac{-GM}{2R^3}$$

$$\text{At } r = 0, V' = \frac{-GM}{2R^3} (3R^2) = \frac{-3GM}{2R}$$

$$\text{Clearly, } V' = \frac{3V}{2}$$

11. (1)

An electric dipole contains two equal and opposite charges.

Thus, the net charge, as viewed from outside a surface enclosing the dipole, is zero.

The net flux enclosed by any closed surface is, then, given by

$$\phi = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$= 0$$

12. (4)

The formula to calculate the dipole moment is given by

$$|\vec{p}| = q|\vec{d}| \quad \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the dipole moment.

$$|\vec{p}| = 0.01 \text{ C} \times 0.4 \text{ mm} \times \frac{1 \text{ m}}{10^3 \text{ mm}}$$

$$= 4.0 \times 10^{-6} \text{ C} \cdot \text{m}$$

The formula to calculate the torque on the dipole in the electric field is given by

$$|\vec{\tau}| = |\vec{p} \times \vec{E}| \quad \dots (2)$$

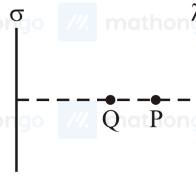
Substitute the values of the known parameters into equation (2) to calculate the required torque.

$$|\vec{\tau}| = 4.0 \times 10^{-6} \text{ C} \cdot \text{m} \times 10 \text{ dyne} \cdot \text{C}^{-1} \times \frac{1 \text{ N}}{10^5 \text{ dyne}} \times \sin 30^\circ$$

$$= 2 \times 10^{-10} \text{ N m}$$

13. (6)

Let's consider the following diagram:



The formula to calculate the electric field due to an infinitely long charged sheet is given by

$$E_s = \frac{\sigma}{2\epsilon_0} \quad \dots(1)$$

The formula to calculate the electric field due to an infinitely long line charge is given by

$$E_l = \frac{\lambda}{2\pi\epsilon_0 r} \quad \dots(2)$$

Using equations (1) and (2), the electric field at P can be calculated as follows:

$$\begin{aligned} E_P &= \left| \frac{\sigma}{2\epsilon_0} - \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{\frac{3}{\pi}} \right| \\ &= \left| \frac{\sigma}{2\epsilon_0} - \frac{\lambda}{6\epsilon_0} \right| \\ &= \frac{\sigma}{6\epsilon_0} \quad \dots(3) \end{aligned}$$

Similarly, the electric field at Q is given by

$$\begin{aligned} E_Q &= \left| \frac{\sigma}{2\epsilon_0} - \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{\frac{4}{\pi}} \right| \\ &= \left| \frac{\sigma}{2\epsilon_0} - \frac{\lambda}{8\epsilon_0} \right| \\ &= \frac{\sigma}{4\epsilon_0} \quad \dots(4) \end{aligned}$$

Divide equation (3) by equation (4) to obtain the required ratio.

$$\begin{aligned} \frac{E_P}{E_Q} &= \frac{\frac{\sigma}{6\epsilon_0}}{\frac{\sigma}{4\epsilon_0}} \\ &= \frac{4}{6} \end{aligned}$$

Hence, $a = 6$.

14. (3)

Let the charges be $x \mu\text{C}$, $(q - x) \mu\text{C}$, where $q = 10 \mu\text{C}$

The force between them is

$$F = \frac{Kx(q-x)}{r^2}$$

For the force to be maximum,

$$\frac{dF}{dx} = 0$$

$$\Rightarrow \frac{dF}{dx} = \frac{K(q-2x)}{r^2} = 0$$

$$\Rightarrow x = \frac{q}{2} = 5 \mu\text{C}$$

15. (5440)

The net force on the given charge will be the resultant of the forces due to the other two charges.

Hence, the net force (F_n) can be calculated as follows:

$$\begin{aligned} F_n &= \frac{1}{4\pi\epsilon_0} \frac{4q^2}{\left(\frac{R}{4}\right)^2} - \frac{1}{4\pi\epsilon_0} \frac{2q^2}{\left(\frac{3R}{4}\right)^2} \\ &= 9 \times 10^9 \times \frac{16q^2}{R^2} \left[4 - \frac{2}{9} \right] \\ &= 9 \times 10^9 \times \frac{16 \times (2 \times 10^{-6} \text{ C})^2}{(0.02 \text{ m})^2} \times \frac{34}{9} \\ &= 5440 \text{ N} \end{aligned}$$

16. (3)                           

An electric dipole is a system of two charges of equal and opposite magnitude placed at a short distance.

The dipole moment is, $p = 2ql$.

The electric field due to the dipole is,

$$E = \frac{2kp}{r^3}$$

Clearly, the electric field varies as $\frac{1}{r^3}$.                    